Tarun Karnati

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Toronto,ON

CAREER OBJECTIVE

Aspiring full-stack developer passionate about building scalable applications and intelligent, data-driven systems. Experienced in developing end-to-end solutions across frontend, backend, databases, and cloud platforms. Driven to leverage AI and emerging technologies to solve real-world problems and deliver impactful, user-focused products.

TECHNICAL SKILLS

Languages: C, Java, Python, JavaScript, HTML5, CSS3, SQL, Bash

Frameworks & APIs: Flask, Spring Boot, Node.js, Express.js, RESTful APIs

Databases: MongoDB, MySQL

AI: MCP, LLM Integration, AI Chatbot Development, Fine-Tuning, RAG, LangChain, OpenAI API, Hugging Face

Cloud & DevOps: Microsoft Azure, Docker, Render, Netlify, Railway

Tools: Git, Postman

Machine Learning:: Supervised & Unsupervised Learning, Regression, Classification, Clustering, Feature Engineering, Model Evaluation (Precision, Recall, F1-score), scikit-learn, TensorFlow, Pandas, NumPy

EDUCATION

INFORMATION TECHNOLOGY SOLUTIONS, 2025 – 2026 | TORONTO
HUMBER COLLEGE INSTITUTE OF TECHNOLOGY

BACHELOR OF TECHNOLOGY - DATA SCIENCE 2020 – 2024 | INDIA
SRK INSTITUTE OF TECHNOLOGY

PROJECTS

ORCA AI – MCP-BASED AI DEVOPS AUTOMATION PLATFORM, *Python, Flask, MCP, Multi-LLM*

- Built an autonomous development system using the **Model Context Protocol (MCP)** to orchestrate multiple tools and AI agents.
- Integrated MCP clients for browser testing, deployment, GitHub automation, and shell execution in a unified workflow.
- Developed multi-model AI orchestration using OpenAI, Anthropic, and Google with intelligent fallback handling.
- Implemented self-healing pipeline where AI detects errors and automatically fixes generated code.
- Enabled end-to-end automation from prompt → code generation → deployment on Railway with live validation.

PERKPAL – PERSONALIZED DISCOUNTS WEB APP, *Full Stack Project | HTML, CSS, JavaScript, Spring Boot, MongoDB*

- Built a full stack web app delivering targeted offers for students, seniors, LGBTQ+, and other communities.
- Created a responsive frontend using HTML, CSS, and JavaScript with interactive forms and validation.
- Built backend APIs using Spring Boot to handle user registration, login, and offer management.
- Used MongoDB to store user and offer data, allowing fast access and filtering.
- Applied logic to show community-specific discounts (e.g., for students, seniors, LGBTQ+) based on user input.

PUBLIC RELATIONSHIPS USING NLP AND ML, *NLP, Machine Learning, Logistic Regression, Naive Bayes, Sentiment Analysis*

- Developed an NLP-based sentiment analysis model to classify reviews and help businesses understand customer opinion.
- Used Logistic Regression and Naive Bayes to deliver accurate predictions for data-driven decisions.
- Applied text preprocessing techniques to improve model performance and classification quality.
- Analyzed customer feedback to identify trends and enhance customer satisfaction and brand reputation.
- Built a scalable solution for real-time sentiment monitoring with minimal system requirements.

PROFESSIONAL EXPERIENCE

AWS CLOUD VIRTUAL INTERNSHIP

- Gained hands-on exposure to core AWS services such as EC2, S3, and IAM, building a foundational understanding of cloud computing.
- Learned key concepts like deployment, storage, security, and scalability in cloud environments through practical tasks.
- Worked on basic cloud-based activities to understand resource provisioning and architecture design principles.

DATA ANALYTICS VIRTUAL INTERSHIP

- Performed data cleaning, preprocessing, and basic analysis using tools like Excel and Python.
- Created visualizations and reports to identify trends and extract meaningful insights from datasets.
- Applied fundamental data analytics techniques to support simple data-driven decision-making scenarios.

HACKATHONS

ALBERTA AI GATEWAY – AI & XR POWERED SMART TOURISM PLATFORM, *2nd Prize Hackathon Winner*

- Built and led a full-stack AI platform (React, Node.js, Express) in a 3-day hackathon, connecting Alberta globally across tourism, trade, culture, and education.

- Integrated Whisper (speech-to-text), Gemini/GPT (text & image generation), and Text-to-Speech to create fully voice-enabled, multi-day travel itineraries.
- Connected Ticketmaster API (real-time events) and OpenWeather API (live weather) to dynamically enrich trip plans.
- Developed immersive XR features using interactive 3D (.glb) models and a multilingual AI chatbot for enhanced user engagement.
- Designed a secure backend with JWT authentication, saved user plans, protected API keys, and cloud deployment.
- Deployed live: https://alberta-ai-gateway.vercel.app/ demonstrating full functionality and real-world impact.

CERTIFICATES

- INTRODUCTION TO MODEL CONTEXT PROTOCOL FROM ANTHROPIC
- MICROSOFT AZ900: AZURE FUNDAMENTALS FROM MICROSOFT
- GENERATIVE AI: WORKING WITH LARGE LANGUAGE MODELS FROM NASBA AND LINKEDIN LEARNING
- DOCKER FOUNDATIONS PROFESSIONAL CERTIFICATE – LINKEDIN LEARNING & DOCKER
- CAREER ESSENTIALS IN SOFTWARE DEVELOPMENT BY MICROSOFT AND LINKEDIN

ACHIEVEMENTS AND AWARDS

DEAN'S HONOUR LIST – HUMBER COLLEGE, *AWARDED FOR ACADEMIC EXCELLENCE IN SEMESTER 1 (2025)*

DEAN'S HONOUR LIST – HUMBER COLLEGE, *AWARDED FOR ACADEMIC EXCELLENCE IN SEMESTER 2 (2025)*

PAPER PUBLICATIONS

PUBLIC RELATIONSHIPS USING NLP AND MACHINE LEARNING, *(IJIRT 2024)*
Published research in the *International Journal of Creative Research Thoughts (IJCRT)* on leveraging NLP and machine learning techniques, including Logistic Regression and Naive Bayes, to analyze online product reviews. The study focused on extracting customer sentiment, identifying negative feedback, and providing insights to help businesses improve decisions and understand public opinion.